Trainings

General Information

Cummins Inc. requires that diesel fuel meeting the specifications in Table 1: Cummins Inc. Required Fuel Properties in this service manual be used in Cummins® Marine engines. However, the possibility exists that fuel of this quality may **not** be readily available in certain Marine markets. The International Standards Organization (ISO) has defined specifications for fuels called Products from Petroleum, Synthetic and Renewable Sources – Fuels (class F) – Specifications of Marine Fuels. This specification defines seven distinct fuels; DMX, DMA, DFA, DMB, DFB, DMZ and DFZ. The F notation in three of these fuels signifies that the fuel could contain up to 7 percent biodiesel, but fuel properties are similar to the M designations. For example: DMA and DFA have similar fuel properties, but DFA contains up to 7 percent biodiesel. The characteristics of these fuels are presented in Table 12: Marine Fuel Characteristics.

Except as specified in Table 11: Acceptable Use Of Category ISO 8217 Fuels, Cummins Inc. does **not** recommend the use of fuels meeting the specifications in Table 12, because some characteristics of these fuels do **not** meet the required diesel fuel specifications in Table 1 of this service manual. Some DMX, DMA/DFA, and DMB/DFB or DMZ/DFZ fuels may meet the specifications listed in Table 3: Contingency Diesel Fuel Specifications, and if so, could be used as such. Additionally, in some areas, such as the European Union Territory, the sulfur content has been limited to 0.2 mass percent (2000 ppm) or less for all category ISO 8217 fuels. Low sulfur Marine fuel is **not** available in all markets. It is the user's responsibility to select the correct fuel and make sure that the fuel properties satisfy Cummins Inc. requirements.

Table 11, Acceptable Use of Category ISO 8217 Fuels

Product Series	Displacement (Liters)	Products	Emissions Certification	ISO 8217 Fuel			
- - -	- - -	- - -	- - -	DMA/DFA	DMB/DFB	DMZ/DFZ	DMX
QSK	95	QSK95 CM2350 K128	IMO Tier II	*Yes	No	No	No
KTA	19, 38, 50	All Marine Engines	IMO Tier II	*Yes	No	No	No

Table 11, Acceptable Use of Category ISO 8217 Fuels							
Product Series	**Displacement (Liters)**	**Products**	**Emissions Certification**	**ISO 8217 Fuel**			
QSK	19, 38, 50, 60	All Marine Engines	IMO Tier II	*Yes	No	No	No
X	15	All Marine Engines	IMO Tier II	*Yes	No	No	No
QSB	4.5, 5.9, 6.7, 7	All Marine Engines	IMO Tier II	*Yes	No	No	No
QSL	9	All Marine Engines	IMO Tier II	*Yes	No	No	No
QSM	11	All Marine Engines	IMO Tier II	*Yes	No	No	No

*ISO 8217 DMA fuel sulfur should meet Cummins requirements in the appropriate engine owner's manual. Engines with lower tier emissions certifications are approved as well.

Fuels meeting the specifications in Table 12 for DMA/DFA are acceptable for use **only** in engines specified in Table 11.

*ISO 8217 DMA fuel sulfur should meet Cummins requirements in the appropriate engine owners manual. Engines with lower tier emissions are approved as well.

Table 12, Marine Fuel Characteristics						
Characteristics	**Limit**	**Category ISO 8217**				**Test Method Reference**
- - -	**- - -**	**DMA /DFA**	**DMX**	**DMB /DFB**	**DM Z /DFZ**	**- --**
Appearance	N/A	Bright and Clear				N/A
Density at 15°C [59°F], kg/m ^{3}	Maximum	(1)890	----	900	890	ISO 3675 or ISO 12185

Table 12, Marine Fuel Characteristics						
Characteristics	**Limit**	**Category ISO 8217**				**Test Method Reference**
- - -	- - -	**DMA /DFA**	**DMX**	**DMB /DFB**	**DM Z /DFZ**	- --
Viscosity at 40°C [104°F], centistokes	Minimum	2.0	1.45	2.0	3.0	ISO 3104
	Maximum	6.0	5.5	11.0	6.0	ISO 3104
Flash Point, °C	Minimum	43	60	60	60	ISO 2719
Pour Point (upper), °C(2) — Winter Quality	Maximum	-6	----	0	-6	ISO 3016
Pour Point (upper), °C(2) — Summer Quality	Maximum	0	----	6	0	ISO 3016
Cloud Point, °C	Maximum	Report	-16	(4)Report	(4)Report	ISO 3015
Cold filter plugging point, °C		(4) Report				EN 116, or EN 16329
Sulfur, mass percent (3)	Maximum	(5)Per Statutory Requirements and Cummins Use Requirements				ISO 8754
Hydrogen Sulfide, mg/kg	Maximum	2.00				IP 570
Acid Number, mg KOH/g	Maximum	0.5				ASTM D664
Cetane Index	Minimum	40 (DMA)	45	35 (DMB)	40 (DMZ)	
Cetane Number	Minimum	40 (DFA)	40	35 (DFB)	40 (DFZ)	ISO 5165
Carbon Residue (micro method), mass percent 10 percent (volume) distillation, bottoms	Maximum	0.30	0.30	- - -	- - -	ISO 10370
Carbon Residue (micro method), mass percent	Maximum	- - -	- - -	0.30	2.50	ISO 10370

Table 12, Marine Fuel Characteristics						
Characteristics	**Limit**	**Category ISO 8217**				**Test Method Reference**
- - -	**- - -**	**DMA /DFA**	**DMX**	**DMB /DFB**	**DM Z /DFZ**	**- --**
Ash, mass percent	Maximum	0.01	0.01	0.01	0.051	ISO 6245
Sediment, mass percent	Maximum	- - -	- - -	0.07	- - -	ISO 3735
Total Existent Sediment, mass percent	Maximum	- - -	- - -	0.10	- - -	ISO 10307-1
Water, volume percent	Maximum	- - -	- - -	0.3	- - -	ISO 14597
Vanadium, mg/kg	Maximum	- - -	- - -	- - -	100	ISO 14597
Aluminum plus silicon, mg/kg	Maximum	- - -	- - -	- - -	25	ISO 10478
Oxidation Stability, g/m^3	Maximum	20 (DMA)	25	25 (DMB)	25 (DMZ)	ISO 12205
Oxidation Stability, h	Minimum	8.0 (DFA)	---	8.0 (DFB)	8.0 (DFZ)	EN 15751
Fatty Acid methyl ester (FAME)	Maximum	7.0% (DFA)	---	---	---	ASTM D7963, or EN 14078/ ASTM D7371
Lubricity, corrected wear scar diameter (WSD) at 60°C, micrometers	Maximum	520				ISO 12156-1

Table 12, Marine Fuel Characteristics						
Characteristics	**Limit**	**Category ISO 8217**				**Test Method Reference**
- - -	**- - -**	**DMA /DFA**	**DMX**	**DMB /DFB**	**DM Z /DFZ**	**- --**

(1) In some geographical areas, there may be a maximum density limit.
(2) Purchasers are recommended to make sure that this pour point is suitable for the equipment on board, especially if the vessel operates in both the northern and southern hemispheres.1.0 mass percent = 10,000 ppm.
(3) Report Values to purchaser but are not required by this specification.
(4) Fuel Sulfur should conform to statutory limits in the geographic area of operation and Cummins requirements per
engine owners manual.

Warranty and the use of Marine Distillate Oils in Cummins® Engines

Cummins Inc. Engine Warranty covers failures that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by the use of MDO fuel are **not** considered to be defects in material or workmanship, and are **not** covered under Cummins Inc. engine warranty.

Last Modified: 02-Dec-2024